Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Manuel Martins Dairy L.P.

Physical address of dairy:

Number and Street	City	County	Zip Code
10043 Milnes	Modesto	Stanislaus	95307

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/09/1978

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X015-X001-X020-XXXX

B. OPERATORS

Martin, David

Operator name: Martin, David

Telephone no.: (209) 529-8133 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3213 Oakdale Waterford HWY	Oakdale	CA	95361

This operator is responsible for paying permit fees.

C. OWNERS

Martin, David

Legal owner name: Martin, David

Telephone no.: (209) 529-8133 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3213 Oakdale Waterford HWY	Oakdale	CA	95361

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	25	0	0	0
Number under roof	820	100	0	0	135	85
Maximum number	820	100	25	0	135	85
Average number	820	100	25	0	135	85
Avg live weight (lbs)	1,400	1,450	950	0		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,321.72 *tons per reporting period*

Total nitrogen from manure: 304,359.67 *lbs per reporting period*

After ammonia losses (30% loss applied): 213,051.77 *lbs per reporting period*

Total phosphorus from manure: 49,700.58 *lbs per reporting period*

Total potassium from manure: 155,960.09 *lbs per reporting period*

Total salt from manure: 409,092.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,780,000 *gallons*

Total nitrogen generated: 48,272.09 *lbs*

Total phosphorus generated: 6,417.75 *lbs*

Total potassium generated: 84,913.91 *lbs*

Total salt generated: 412,126.17 *lbs*

	8,280,000	*gallons applied*
+	1,500,000	*gallons exported*
-	0	*gallons imported*
=	9,780,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
OID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	Corral solids	2,460.00 *ton*	Dry-weight	53.3		22,800.00	5,030.00	44,500.00		29.60

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
05/22/2023	Process wastewater	1,500,000.00 *gal*	507.00	255.00	0.00	0.00	70.80	913.00		4,340

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	52,386.19	11,557.13	102,244.98	680,101.44
Process wastewater	6,346.37	886.24	11,428.48	54,325.95
Total exports for all materials	58,732.56	12,443.37	113,673.46	734,427.39

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	23	23	2	process wastewater	X015-X001-X020-XXXX
Field 2	21	21	2	process wastewater	X015-X001-X020-XXXX
Field 3	28	28	2	process wastewater	X015-X001-X020-XXXX
Field 4	18	18	1	process wastewater	X015-X001-X020-XXXX
Totals for areas that were used for application	90	90	7		
Totals for areas that were not used for application					
Land application area totals	90	90	7		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/02/2022: Ryegrass

Crop: Ryegrass Acres planted: 23 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	329.59 *ton*	As-is		68.0	10,700.00	1,940.00	10,600.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	255.00	37.50	246.00	0.00
Total actual harvest content	14.33	306.66	55.60	303.80	990.49

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	354.20 *ton*	As-is		51.8	5,590.00	1,220.00	4,450.00		3.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	15.40	172.17	37.58	137.06	454.28

Field 2

Field name: Field 2

11/02/2022: Ryegrass

Crop: Ryegrass Acres planted: 21 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	300.93 *ton*	As-is		68.6	10,700.00	1,940.00	10,600.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	255.00	37.50	246.00	0.00
Total actual harvest content	14.33	306.66	55.60	303.80	971.92

06/16/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 21 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	403.20 *ton*	As-is		70.0	4,500.00	850.00	10,800.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	220.00	34.00	240.00	0.00
Total actual harvest content	19.20	172.80	32.64	414.72	1,347.84

Field 3

Field name: Field 3

11/02/2022: Ryegrass

Crop: Ryegrass Acres planted: 28 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	401.24 *ton*	As-is		64.0	7,630.00	1,540.00	5,770.00		9.06

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	255.00	37.50	246.00	0.00
Total actual harvest content	14.33	218.68	44.14	165.37	934.77

Field 3

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	476.70 *ton*	As-is		53.6	5,990.00	1,110.00	4,160.00		4.12

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	17.03	203.96	37.80	141.65	650.93

Field 4

Field name: Field 4

01/01/2023: Pasture

Crop: Pasture Acres planted: 18 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	324.00 *ton*	As-is		68.3	6,260.00	800.00	5,620.00		7.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	331.20	46.80	340.20	0.00
Total actual harvest content	18.00	225.36	28.80	202.32	866.17

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/02/2022: Ryegrass

Field name: Field 1

Crop: Ryegrass Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	139.34	20.30	237.72	1,115.47	504,000.00 *gal*
Application event totals		139.34	20.30	237.72	1,115.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	159.25	23.20	271.68	1,274.83	576,000.00 *gal*
Application event totals		159.25	23.20	271.68	1,274.83	

Field 1 - 06/02/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	218.54	30.52	393.54	1,870.70	1,188,000.00 *gal*
OID Canal	Surface water	0.82	0.00	0.00	42.03	2,227,500.00 *gal*
Application event totals		219.35	30.52	393.54	1,912.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.14	0.00	0.00	58.58	3,105,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

Field 1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.19	0.00	0.00	61.13	3,240,000.00 *gal*
Application event totals		1.19	0.00	0.00	61.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.14	0.00	0.00	58.58	3,105,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.99	0.00	0.00	50.94	2,700,000.00 *gal*
Application event totals		0.99	0.00	0.00	50.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.89	0.00	0.00	45.85	2,430,000.00 *gal*
Application event totals		0.89	0.00	0.00	45.85	

Field 2 - 11/02/2022: Ryegrass

Field name: Field 2

Crop: Ryegrass

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	141.71	20.64	241.77	1,134.44	468,000.00 *gal*
Application event totals		141.71	20.64	241.77	1,134.44	

Field 2 - 11/02/2022: Ryegrass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	152.61	22.23	260.36	1,221.71	504,000.00 *gal*
Application event totals		152.61	22.23	260.36	1,221.71	

Field 2 - 06/16/2023: Sudangrass, silage

Field name: Field 2

Crop: Sudangrass, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.14	0.00	0.00	58.58	2,835,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.03	0.00	0.00	53.00	2,565,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	220.94	17.93	386.83	2,149.86	936,000.00 *gal*
OID Canal	Surface water	0.70	0.00	0.00	36.26	1,755,000.00 *gal*
Application event totals		221.64	17.93	386.83	2,186.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.03	0.00	0.00	53.00	2,565,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.00	

Field 2 - 06/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.03	0.00	0.00	53.00	2,565,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.00	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.98	0.00	0.00	50.21	2,430,000.00 *gal*
Application event totals		0.98	0.00	0.00	50.21	

Field 3 - 11/02/2022: Ryegrass

Field name: Field 3

Crop: Ryegrass Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	155.34	22.63	265.01	1,243.52	684,000.00 *gal*
Application event totals		155.34	22.63	265.01	1,243.52	

04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.67	16.71	215.51	1,024.43	792,000.00 *gal*
Application event totals		119.67	16.71	215.51	1,024.43	

Field 3 - 06/02/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage Plant date: 06/02/2023

Field 3 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	228.47	31.90	411.43	1,955.73	1,512,000.00 *gal*
OID Canal	Surface water	0.85	0.00	0.00	43.94	2,835,000.00 *gal*
Application event totals		229.32	31.90	411.43	1,999.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.14	0.00	0.00	58.58	3,780,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.10	0.00	0.00	56.49	3,645,000.00 *gal*
Application event totals		1.10	0.00	0.00	56.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.06	0.00	0.00	54.40	3,510,000.00 *gal*
Application event totals		1.06	0.00	0.00	54.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.02	0.00	0.00	52.31	3,375,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	1.02	0.00	0.00	52.31	3,375,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.31	

Field 4 - 01/01/2023: Pasture

Field name: Field 4

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.88	0.00	0.00	45.56	1,890,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	135.39	18.91	243.81	1,158.95	576,000.00 *gal*
OID Canal	Surface water	0.51	0.00	0.00	26.04	1,080,000.00 *gal*
Application event totals		135.89	18.91	243.81	1,184.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.88	0.00	0.00	45.56	1,890,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.88	0.00	0.00	45.56	1,890,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.85	0.00	0.00	43.94	1,822,500.00 *gal*
Application event totals		0.85	0.00	0.00	43.94	

Field 4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.92	0.00	0.00	47.19	1,957,500.00 *gal*
Application event totals		0.92	0.00	0.00	47.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	148.71	12.07	260.36	1,447.02	540,000.00 *gal*
OID Canal	Surface water	0.92	0.00	0.00	47.19	1,957,500.00 *gal*
Application event totals		149.62	12.07	260.36	1,494.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.92	0.00	0.00	47.19	1,957,500.00 *gal*
Application event totals		0.92	0.00	0.00	47.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.88	0.00	0.00	45.56	1,890,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.85	0.00	0.00	43.94	1,822,500.00 *gal*
Application event totals		0.85	0.00	0.00	43.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.47	0.00	0.00	24.41	1,012,500.00 *gal*
Application event totals		0.47	0.00	0.00	24.41	

Field 4 - 01/01/2023: Pasture

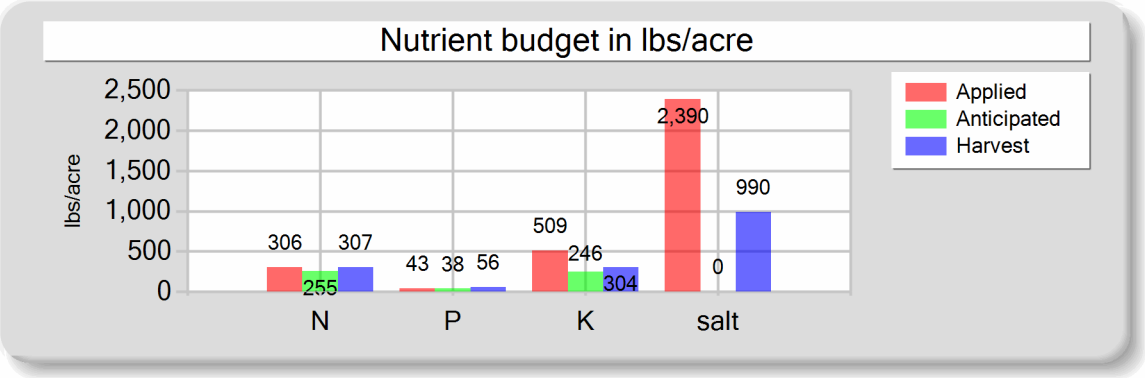
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Canal	Surface water	0.88	0.00	0.00	45.56	1,890,000.00 *gal*
Application event totals		0.88	0.00	0.00	45.56	

B. NUTRIENT BUDGET

Field 1 - 11/02/2022: Ryegrass

Field name: Field 1 Crop: Ryegrass Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	298.59	43.50	509.41	2,390.30
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.59	43.50	509.41	2,390.30
Anticipated crop nutrient removal	255.00	37.50	246.00	0.00
Actual crop nutrient removal	306.66	55.60	303.80	990.49
Nutrient balance	-1.07	-12.10	205.61	1,399.81
Applied to removed ratio	1.00	0.78	1.68	2.41

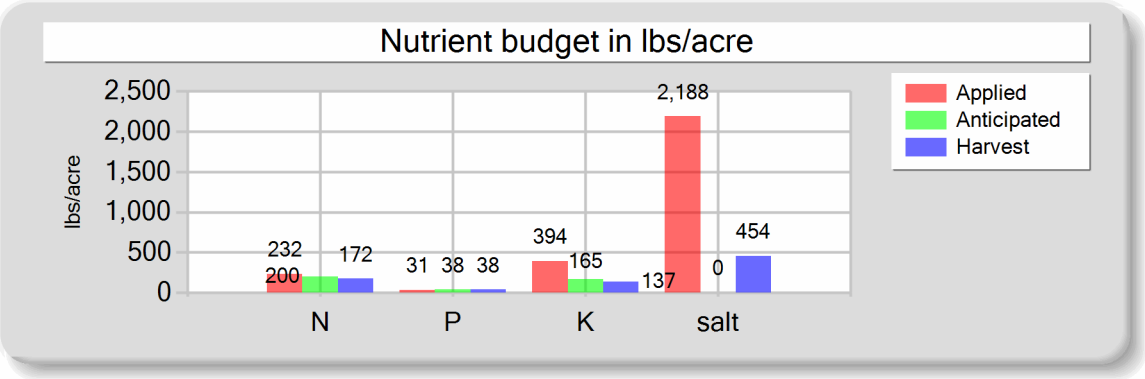
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,080,000.00 *gallons*
39.77 *acre-inches*
1.73 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1 - 06/02/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	218.54	30.52	393.54	1,870.70
Fresh water	6.16	0.00	0.00	317.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.70	30.52	393.54	2,187.81
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	172.17	37.58	137.06	454.28
Nutrient balance	59.52	-7.06	256.48	1,733.53
Applied to removed ratio	1.35	0.81	2.87	4.82

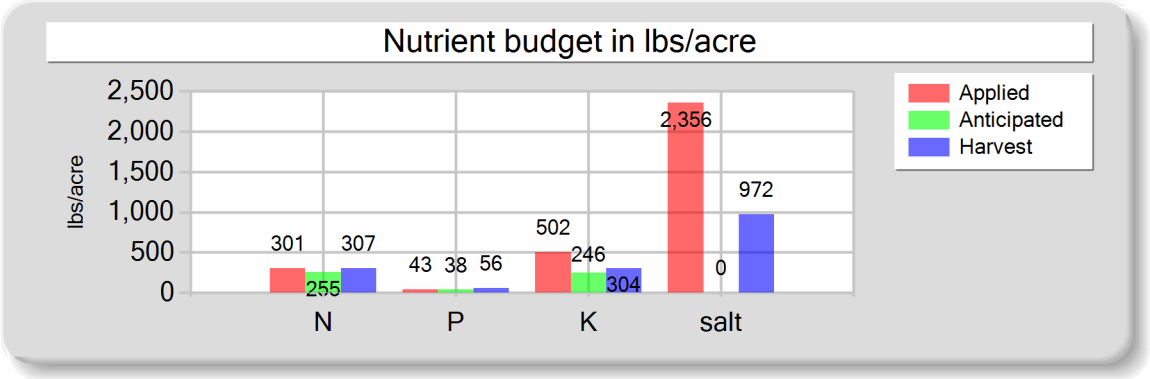
Fresh water applied	
16,807,500.00	*gallons*
618.96	*acre-inches*
26.91	*inches/acre*

Process wastewater applied	
1,188,000.00	*gallons*
43.75	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/02/2022: Ryegrass

Field name: Field 2 Crop: Ryegrass Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.33	42.87	502.13	2,356.15
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.33	42.87	502.13	2,356.15
Anticipated crop nutrient removal	255.00	37.50	246.00	0.00
Actual crop nutrient removal	306.66	55.60	303.80	971.92
Nutrient balance	-5.34	-12.73	198.33	1,384.23
Applied to removed ratio	0.98	0.77	1.65	2.42

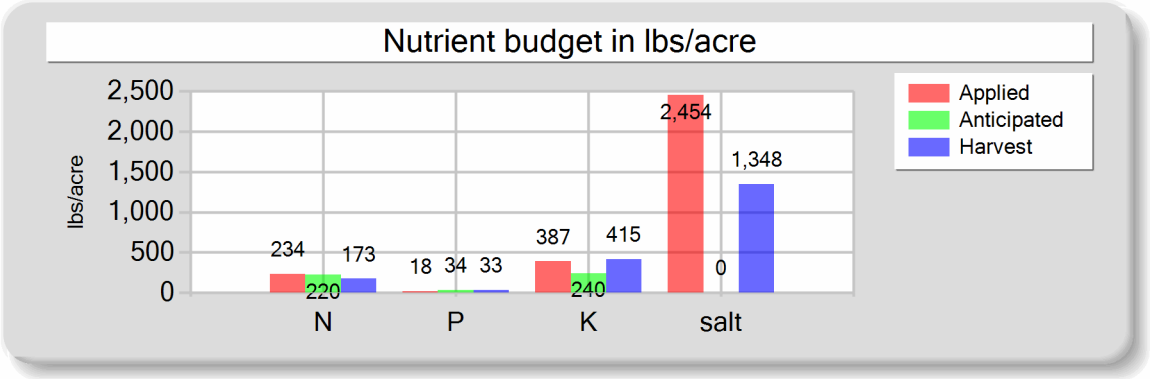
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
972,000.00	*gallons*
35.80	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/16/2023: Sudangrass, silage

Field name: Field 2 Crop: Sudangrass, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.94	17.93	386.83	2,149.86
Fresh water	5.91	0.00	0.00	304.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.84	17.93	386.83	2,453.93
Anticipated crop nutrient removal	220.00	34.00	240.00	0.00
Actual crop nutrient removal	172.80	32.64	414.72	1,347.84
Nutrient balance	61.04	-14.71	-27.89	1,106.09
Applied to removed ratio	1.35	0.55	0.93	1.82

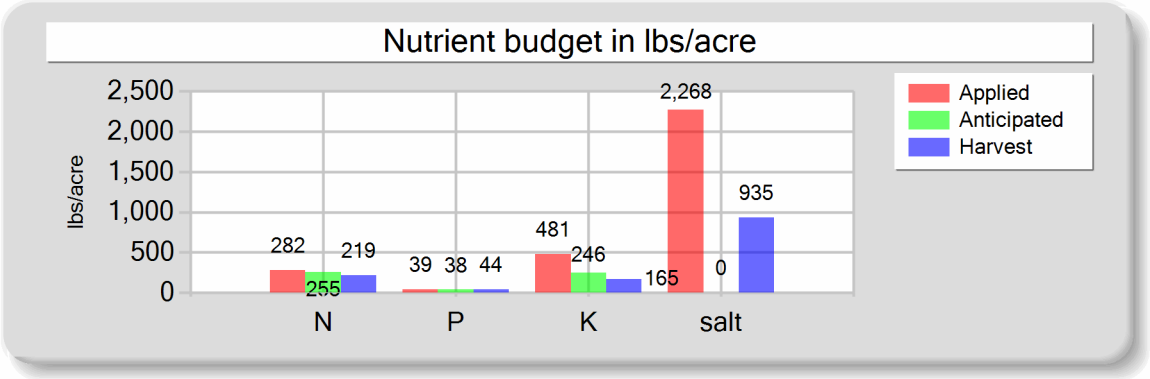
Fresh water applied	
14,715,000.00	*gallons*
541.90	*acre-inches*
25.80	*inches/acre*

Process wastewater applied	
936,000.00	*gallons*
34.47	*acre-inches*
1.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/02/2022: Ryegrass

Field name: Field 3 Crop: Ryegrass Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.01	39.34	480.52	2,267.96
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.01	39.34	480.52	2,267.96
Anticipated crop nutrient removal	255.00	37.50	246.00	0.00
Actual crop nutrient removal	218.68	44.14	165.37	934.77
Nutrient balance	63.34	-4.80	315.15	1,333.18
Applied to removed ratio	1.29	0.89	2.91	2.43

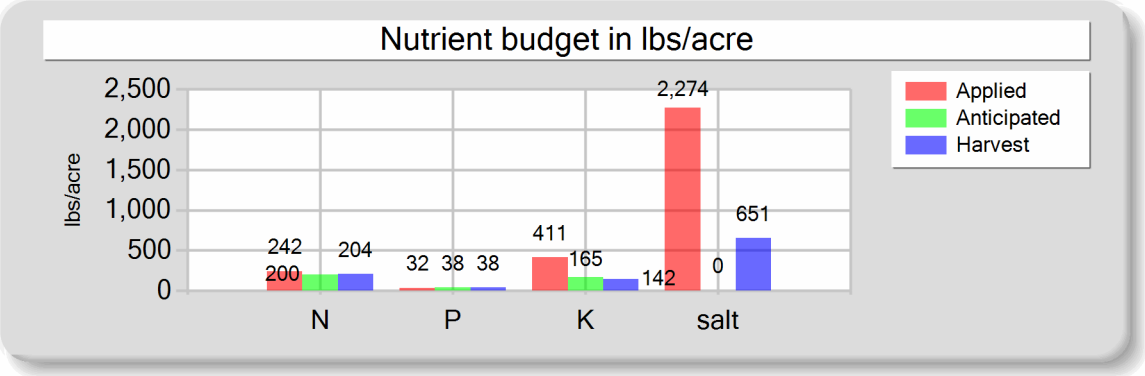
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,476,000.00	*gallons*
54.36	*acre-inches*
1.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/02/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.47	31.90	411.43	1,955.73
Fresh water	6.18	0.00	0.00	318.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.65	31.90	411.43	2,273.75
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	203.96	37.80	141.65	650.93
Nutrient balance	37.69	-5.89	269.78	1,622.82
Applied to removed ratio	1.18	0.84	2.90	3.49

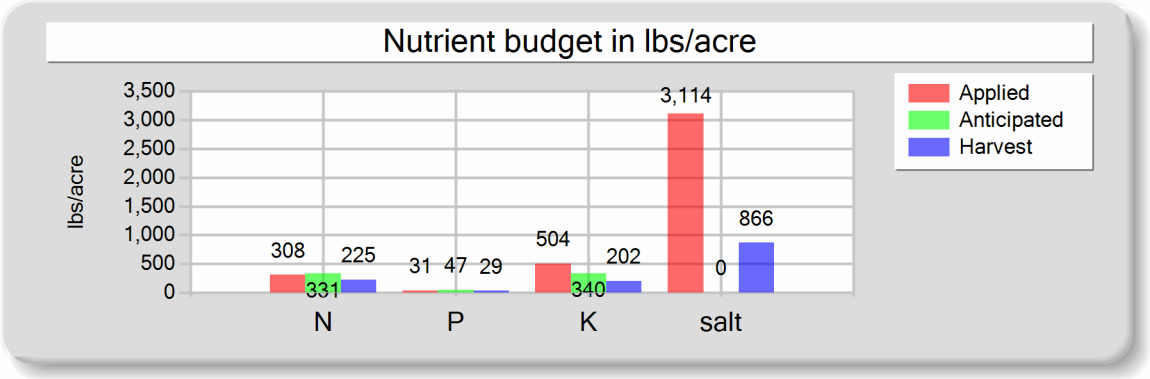
Fresh water applied	
20,520,000.00	*gallons*
755.68	*acre-inches*
26.99	*inches/acre*

Process wastewater applied	
1,512,000.00	*gallons*
55.68	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 01/01/2023: Pasture

Field name: Field 4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	284.10	30.97	504.17	2,605.98
Fresh water	9.86	0.00	0.00	507.71
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	307.96	30.97	504.17	3,113.69
Anticipated crop nutrient removal	331.20	46.80	340.20	0.00
Actual crop nutrient removal	225.36	28.80	202.32	866.17
Nutrient balance	82.60	2.17	301.85	2,247.52
Applied to removed ratio	1.37	1.08	2.49	3.59

Fresh water applied	
21,060,000.00	*gallons*
775.57	*acre-inches*
43.09	*inches/acre*

Process wastewater applied	
1,116,000.00	*gallons*
41.10	*acre-inches*
2.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	4,340.00	30,800.00	20,700.00	6,550.00	3,630.00	2,300.00	1,390.00		41.60
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	1,000.00		0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 04/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	5,410.00	9,370.00							19.50
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,800.00	5,030.00	44,500.00							29.60
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 10/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	3,060.00	10,100.00							28.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

2022 4QWW

Sample and source description: 2022 4QWW

Sample date: 10/26/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	601.00	359.00	0.00	0.00	93.90	1,200.00								6,630.00	3,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								10.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	762.00	456.00	0.00	0.00	111.00	1,300.00	201.00	125.00	131.00	3,800.00	0.00	262.00	358.00	7,960.00	6,100
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	0.50	1.00	0.10	10.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	507.00	255.00	0.00	0.00	70.80	913.00								5,400.00	4,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								10.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	594.00	459.00	0.00	0.00	48.20	1,040.00								7,270.00	5,780
DL	0.50	0.50	0.50	0.50	0.50	0.50								10.00	10

4QWW

Sample and source description: 4QWW

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	809.00	496.00	0.00	0.00	62.40	1,480.00								9,750.00	7,920
DL	0.50	0.50	0.50	0.50	0.50	0.50								10.00	10

C. FRESH WATER ANALYSES

OID Canal

OID Canal

OID Canal

Sample description: OID Canal

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

Field 1

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

Field 2

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

Field 3

Field 3

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00				
DL			4.00				

Field 4

Soil Sample

Sample and source description: Soil Sample

Sample date: 10/18/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/02/2022: Ryegrass

Ryegrass

Sample and source description: Ryegrass

Sample date: 05/02/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,940.00	10,600.00		10.80
DL	100.00	100.00	100.00		0.01

Field 1 - 06/02/2023: Corn, silage

Field 1 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,590.00	1,220.00	4,450.00		3.06
DL	100.00	100.00	100.00		0.01

Field 2 - 11/02/2022: Ryegrass

Ryegrass

Sample and source description: Ryegrass

Sample date: 05/06/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,940.00	10,600.00		10.80
DL	100.00	100.00	100.00		0.01

Field 2 - 06/16/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	850.00	10,800.00		11.70
DL	100.00	100.00	100.00		0.01

Field 3 - 11/02/2022: Ryegrass

Ryegrass

Sample and source description: Ryegrass

Sample date: 05/06/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,630.00	1,540.00	5,770.00		9.06
DL	100.00	100.00	100.00		0.01

Field 3 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,990.00	1,110.00	4,160.00		4.12
DL	100.00	100.00	100.00		0.01

Field 4 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,260.00	800.00	5,620.00		7.59
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

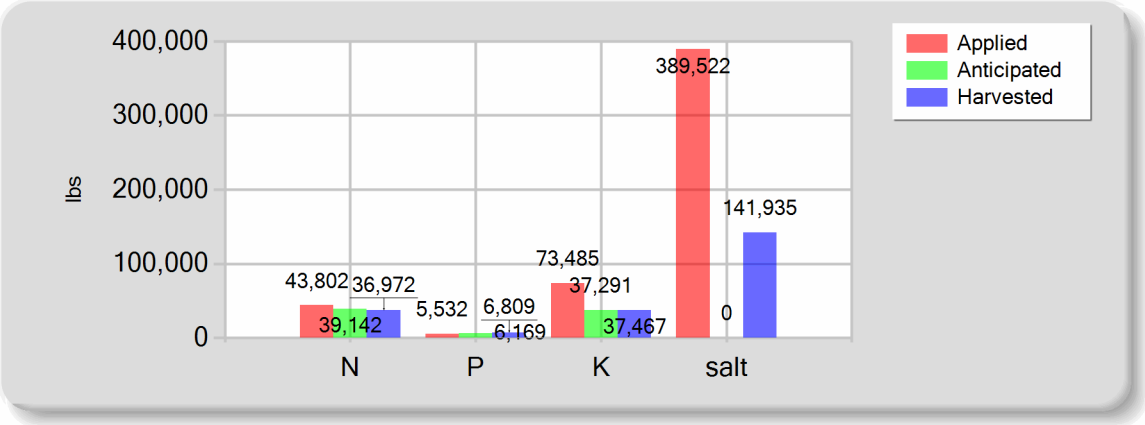
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

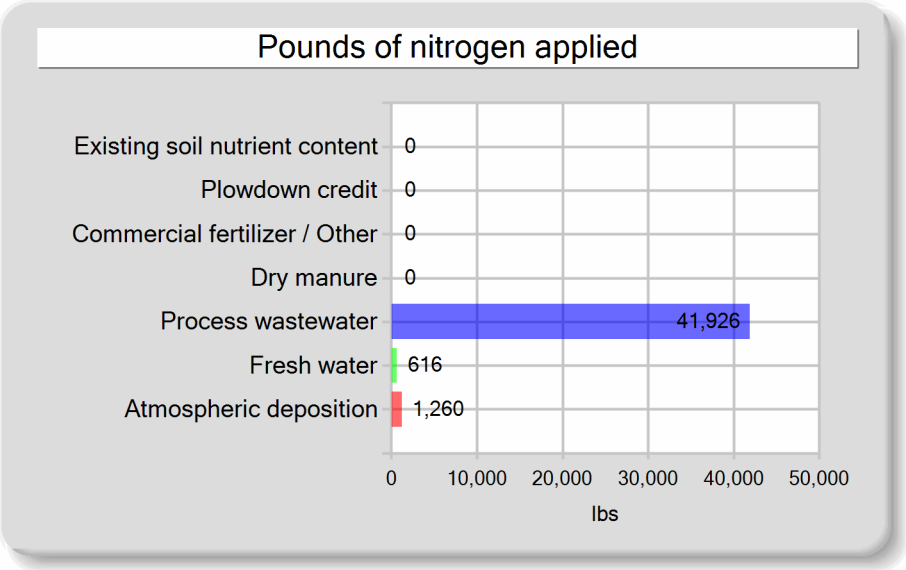
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

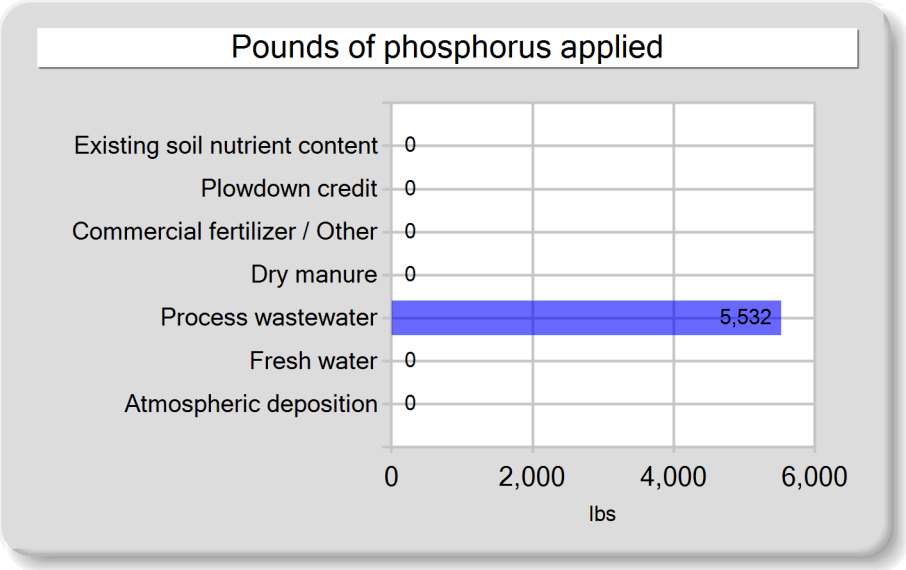
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	41,925.71	5,531.51	73,485.44	357,800.22
Fresh water	616.14	0.00	0.00	31,722.10
Atmospheric deposition	1,260.00	0.00	0.00	0.00
Total nutrients applied	43,801.85	5,531.51	73,485.44	389,522.32
Anticipated crop nutrient removal	39,141.60	6,168.90	37,290.60	0.00
Actual crop nutrient removal	36,972.15	6,808.60	37,466.74	141,935.23
Nutrient balance	6,829.70	-1,277.09	36,018.70	247,587.09
Applied to removed ratio	1.18	0.81	1.96	2.74

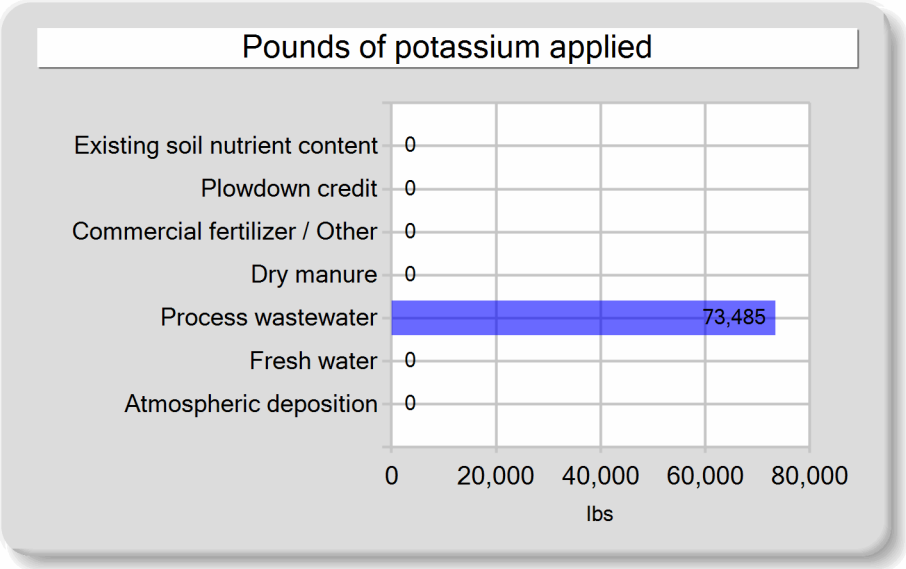
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

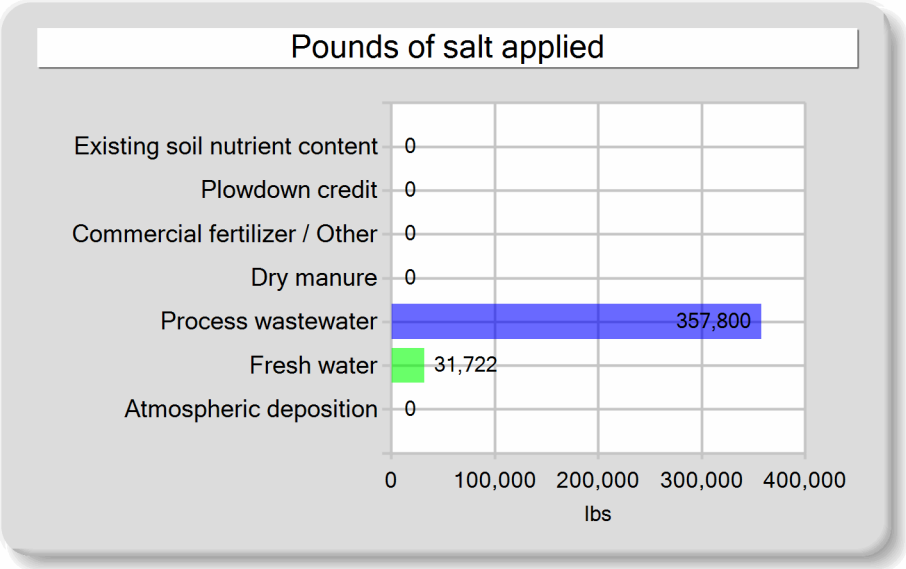


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
David Martin	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-24-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: David Martin

Name of Dairy Facility: Martins Living Trust

Facility Address: 10043 Milnes Rd (Number and Street) Modesto (City) 95307 (Zip Code)

Contact Person Name and Phone Number: David Martin (Name) 209 539-8133 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Viera Bros Farms

Address of Hauling Company /Person: 6960 Crane Rd (Number and Street) Oakdale (City) 95361 (Zip Code)

Contact Person: David Viera (Name) 209 247-3468 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Martins Ranch (Name) 8742 Milnes Rd (Number and Street) Modesto (City) 95357 (Zip Code) 209-380-4814 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

8742 Milnes Rd (Number and Street) Modesto (City) 95357 (Zip Code) 014-033-004 (Assessor's Parcel Number)

Dates Hauled: 5/22/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,500,000 Gallons 2QWW

Method used to determine volume of process wastewater: _____

150 Tanker Loads @ 10,000 gal / Tanker

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Hal M Date: 6-15-24

Hauler's Signature: Dave Vieira Date: 6-15-24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: David Martin

Name of Dairy Facility: Martin's Living Trust

Facility Address: 10043 Milnes Rd (Number and Street) Modesto (City) 95307 (Zip Code)

Contact Person Name and Phone Number: David Martin (Name) 209 529-8133 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: A.L. Barrios

Address of Hauling Company /Person: 9503 Milnes Rd (Number and Street) Modesto (City) 95357 (Zip Code)

Contact Person: A.L. Barrios (Name) 209 204-9260 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above): 209

A.L. Barrios (Name) 9503 Milnes Rd (Number and Street) Modesto, CA (City) 95357 (Zip Code) 204-9260 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

A.L. Barrios 9503 Milnes Rd (Number and Street) Modesto (City) CA 95357 (Zip Code) 014-024-006 (Assessor's Parcel Number)

Dates Hauled: 10/27/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2460 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 53.3 % Moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: 246 Load @ 10 T/Load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mal W Date: 6-15-24

Hauler's Signature: al Barrios Date: 6-15-24

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Manuel Martins Dairy L.P.
3213 Oakdale-Waterford Hwy
Oakdale, CA 95361

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/22/23 1:30 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 22.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3234033A	Electrical Conductivity (EC)	EPA 120.1 and	0.210	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.11	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	135	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.2 (T3234028A)	95.2 (T3234028A)	0.000
Electrical Conductivity (EC)	98.4			2.53
Nitrate Nitrogen (NO3-N)	103	99.9 (W3235002A)	99.9 (W3235002A)	0.060

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Manuel Martins
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F&R Ag Services
Sampled By: Jeanette
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	8/22/23		Milk Barn	Domestic	22.10'		✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jeanette Sanger	F&R		
2nd		DENELE ANALYTICAL, INC.	8-22-23 1:30	

(LABORATORY USE ONLY)

Logged in By:

LABORATORY NUMBER: T3234033